

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `b7d0aa06-0c86-4945-91c6-cbfa74735d90`
- **Timestamp:** 2026-09-05T16:34:09.637010Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** Starting from [2020] 13 S.C.R. 427: which authority does the case it relies on itself rely on, and what does it rely on?
- **Answer:** I do not have enough information to answer this.
- **Health Score:** 0.08
- **Primary Issue:** RETRIEVAL_MISS
- **Claims Supported:** 0 / 1
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
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CORPUS	UNKNOWN	0.5	Corpus relevance is inconclusive.
RETRIEVER	FAIL	0.85	No retrieved chunks provide support, and retrieval scores strongly indicate a lack of useful evidence.
GENERATOR	UNKNOWN	0.5	1 generated claims are unsupported, but retrieval quality is ambiguous.
GROUNDING	PASS	0.9	No claims are contradicted by retrieved evidence.

Efficiency: 0/6 chunks used (0%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 1 (Supported: 0, Partial: 0, Unsupported: 1, Contradicted: 0)
- **Grounding Score:** 0.00
- **Evidence Coverage:** 0.17
- **Average Entailment:** 0.10
- **Latencies:** Retrieval (2460.9ms) | Generation (13918.0ms) | Verification (9103.1ms)

5. RAGAS Metrics

- **Faithfulness:** 0.0000
- **Answer Relevancy:** 0.3596
- **Context Precision:** 0.0000
- **Context Relevancy:** 0.0000
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[167e913d-07a0-47ab-8a72-daf0be9c4327]** (UNSUPPORTED): I do not have enough information to answer this.
- *Evidence (Chunk fa010bb4-6903-474b-912c-073013c906bd):* of the Calendar for 20..... , at the said Session, was convicted of the offence of..... , punishable under section.....of the Bharatiya Nyaya Sanhita, 2023, and was sentenced to..... and thereupon committed to your custody; and whereas by the order of the Court of order of the(a duplicate of which is hereunto annexed) the punishment adjudged by the said sentence has been commuted to the punishment of imprisonment for life;
This is to authorise and require you safely to keep the said..... (prisoner's name) in your custody in the said Jail, as by law is required, until he shall be delivered over by you to the proper authority and custody for the purpose of his undergoing the punishment of imprisonment for life under the said order,
or
if the mitigated sentence is one of imprisonment, say, after the words "custody in the said

Jail", "and there to carry into execution the punishment of imprisonment under the said order according to law".

7. Root Cause Analysis

- **Primary Cause:** RETRIEVAL_MISS
- **Confidence:** 0.88
- **Reasoning:**
 - Starting root cause analysis for trace b7d0aa06-0c86-4945-91c6-cbfa74735d90.
 - Skipping CORPUS: status is UNKNOWN (insufficient evidence).
 - Failure at RETRIEVER: RETRIEVAL_MISS (confidence=0.85). Observation: No retrieved chunks provide support, and retrieval
 - Skipping CHUNKING: no observable state in matrix.
 - Skipping GENERATOR: status is UNKNOWN (insufficient evidence).
 - GROUNDING passed (confidence=0.90).
 - Primary root cause: RETRIEVAL_MISS at RETRIEVER (earliest upstream failure, confidence=0.85).
 - Analysis complete. Primary cause: RETRIEVAL_MISS.

8. Corrective Action Plan

- **[IMMEDIATE] Increase Retrieval Top-K**
 - Description: Increase the number of retrieved documents before providing context to the LLM.
 - Expected Improvement: Higher recall; missing context is caught within the expanded window. (Metric: Verification support)
- **[SHORT_TERM] Implement Hybrid Search (Dense + Sparse)**
 - Description: Add BM25 lexical search and combine with semantic vector search using Reciprocal Rank Fusion.
 - Expected Improvement: Near-perfect recall for exact-match terminology. (Metric: Retrieval hit rate for exact entity queries)
- **[EXPERIMENTAL] Deploy Cross-Encoder Reranker**
 - Description: Pass the Top-N results through a Cross-Encoder to re-score query-document relevance.
 - Expected Improvement: Drastically improved precision at Top-1 and Top-3. (Metric: Precision@3 increases by at least 20%)
- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**
 - Description: Retrieval efficiency is low (0%); consider reducing top-k or tightening the reranker score cutoff.
 - Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

9. Evidence Traceability

- **Primary Cause:** RETRIEVAL_MISS
- **Derived From:** RootCauseAnalysis
- **Pipeline Stage:** RETRIEVAL_MISS
- **Supporting Claims:** 167e913d-07a0-47ab-8a72-daf0be9c4327
- **Supporting Chunks:** fa010bb4-6903-474b-912c-073013c906bd

10. Overall Assessment

- **Major Strength:** GROUNDING is healthy (confidence=90%): No claims are contradicted by retrieved evidence.
- **Major Weakness:** RETRIEVAL_MISS
- **Next Priority:** Increase Retrieval Top-K
- **Recommendation:** Pipeline requires architectural adjustments across multiple stages.

11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework
- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace

artifacts/rag_traces/b7d0aa06-0c86-4945-91c6-cbfa74735d90.json

✓ ClaimSet

artifacts/claim_sets/b7d0aa06-0c86-4945-91c6-cbfa74735d90.json

✓ VerificationResults

artifacts/verification/b7d0aa06-0c86-4945-91c6-cbfa74735d90.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/b7d0aa06-0c86-4945-91c6-cbfa74735d90.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/b7d0aa06-0c86-4945-91c6-cbfa74735d90.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/b7d0aa06-0c86-4945-91c6-cbfa74735d90.json

✓ DiagnosticEvaluationReport

artifacts/reports/b7d0aa06-0c86-4945-91c6-cbfa74735d90.json